

Sports Betting Intelligence Report

Strategy Engine + GitHub/MCP Ecosystem Reference

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Source codebase: **sports-betting-ai-agent** (fly.io deployment)

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Reality check up front: no legitimate strategy produces a "super high" win rate. Professional bettors hit 54-58% long-term. Anyone promising more is running arbitrage, promo-hunting, or lying. The tools and strategies in this report reflect that reality — they focus on **edge discovery, closing line value (CLV), and bankroll math**, not magic.

Part 1 — Your Codebase: Strategy Engine

Analysis of the 9 strategies currently implemented in `src/sports_betting_agent/strategies/`.

Strategy Summary Table

Strategy	Market	Min Edge	Conf Cap	Base Rate
Elo Spread	Spread	2.5%	58%	—
Public Fade	Spread/Total	2.0%	58%	63.8% ATS (NFL)
Price Dispersion	Spread/Total	2.5%	62%	—
MLS Travel Fatigue	Spread	2.0%	56%	3% dog edge
Spread Value	Spread	1.5%	—	—
Total Value	Total	2.5%	—	—
Steam Follow	Spread/Total	2.0%	53%	~57-60%
Total Projection	Total	sport-dep.	57%	—
Reverse Line Mvmt	Spread/Total	2.0%	60%	56-58% ATS
Ensemble (combined)	Spread/Total	positive	72%	varies

Tier 1 — Strongest Historical Signal

1. Reverse Line Movement (RLM) — 56-58% ATS

Highest-conviction single strategy. Requires two signals firing simultaneously: (a) 60%+ of public money on one side, and (b) the line moves the opposite direction (sharps disagree with the crowd). When both align, confidence reaches 60% — the highest cap of any single strategy.

2. Public Fade — 63.8% ATS (NFL dogs, 4-year avg)

Fades teams with 65%+ of public bets. Strongest on NFL underdogs getting <40% of tickets. The market systematically overvalues popular/heavily-bet favorites.

Tier 2 — Quantitative Sharp-Book Signals

3. Steam Follow — ~57-60%

Follows when 3+ sharp books (Pinnacle, Circa, BookMaker) move the same direction. Confidence cap is deliberately conservative (53%) because the edge decays fast — the opportunity closes quickly after sharp action.

4. Elo Spread

In-house Elo power ratings vs. market spread. Min gap 1.5 pts (NFL), 2.0 pts (NBA), 0.5 runs (MLB), 0.4 goals (soccer). Only bets sharp books. Confidence cap 58%.

5. Price Dispersion

Universal cross-sport strategy. Groups lines by (market, selection, line number) across all books. Fires when one book is 2.5%+ cheaper than the 3+ book consensus. Cap 62% — highest of any single strategy.

Tier 3 — Model-Based Projection

6. Total Projection

Blends rolling team scoring averages with park/weather/umpire adjustments, then regresses 50% toward the book line. Per-sport edge thresholds: 1.5 runs (MLB), 6.25 pts (NBA), 0.75 goals (NHL), 3.0 pts (NFL), 4.25 pts (NCAAF), 0.5 goals (MLS), 0.25 (EPL).

7. Spread Value / Total Value

Devigs sharp books to find true fair price, then checks if any soft book offers 1.5-2.5%+ better. Pure cross-book edge.

8. MLS Home Travel Fatigue — ~3% ATS underperformance for travelers

Niche but real: MLS away teams crossing 2+ timezones on <4 days rest cover at a measurable deficit. Bets home spreads at -0.5 or -1 only.

The Ensemble (Combined Engine)

When multiple strategies independently agree on the same selection, confidences are combined via **Bayesian log-odds** (not simple averaging).

Agreeing strategies	Effective confidence
1 strategy at 55%	~55%
2 independent at 55%	~60%
3 independent at 55%	~65%
Hard cap	72%

Any single strategy is marginal. Two or three agreeing independently on the same play is a strong signal — those are the ensemble's 65-72% confidence bets.

Design Rules the System Enforces

Rule	Why
Spreads & totals only — no moneylines	Moneylines have larger vig and are harder to beat consistently
Sharp books only for line-reading	Pinnacle/Circa/BookMaker set the true market; square books lag
Min 2.5% edge before betting	Below this, the vig eats the edge
Market regression 40-60% on projection models	Pure model predictions overfit; shrinking toward line is more robust
Per-strategy confidence caps (53-72%)	Prevents overconfidence; keeps Kelly bet sizing conservative

Part 2 — GitHub & MCP Ecosystem

Top repositories, MCP servers, and tools to extend or benchmark against.

ML-Based Prediction Models

Repo	Stars	What it does
kyleskom/NBA-Machine-Learning-Sports-Betting	1,630	NBA ML with Keras/TensorFlow + GPT analysis
georgedouzas/sports-betting	691	Scikit-learn framework for backtesting strategies
NBA-Betting/NBA_AI	100	NBA predictions with XGBoost/PyTorch
day-mon/sports-betting-ai	101	Deep-learning NBA winner predictor (Rust)
jkrusina/SoccerPredictor	112	LSTM soccer time-series model

Analytical / Math Libraries

Repo	Stars	What it does
nautechsystems/nautilus_trader	22,137	Rust event-driven engine (supports sports betting)
martineastwood/penaltyblog	157	Poisson / Dixon-Coles / Elo / Pi-ratings for football
sedemmler/WagerBrain	300	Essential betting math: Kelly, vig removal, EV
gotoConversion/goto_conversion	108	Shin / power devigging (Kaggle gold-medal winner)

Arbitrage & Line-Shopping

Repo	Stars	What it does
pretrehr/Sports-betting	503	Multi-bookmaker arb (Pinnacle, Betfair, Bwin, ...)
personal-coding/Live-Sports-Arbitrage-Bet-Finder	275	Live arb across FanDuel / DK / William Hill
ryankrumenacker/sports-betting-arbitrage-project	195	Statistical arbitrage notebooks
daankoning/ArbitrageFinder	92	Simple Python arb detector
carterlasalle/SportsArbFinder	10	Odds-API-powered multi-region arb tool

Kelly / Bankroll

Repo	Stars	What it does
chrisgillam/polymarket_gambot	24	Pinnacle vs Polymarket probabilistic + Kelly sizing bot

Data Scrapers & APIs

Repo	Stars	What it does
jordantete/OddsHarvester	154	Playwright scraper for oddsportal.com

Repo	Stars	What it does
cvidan/bet365-scraper	154	Selenium-based Bet365 scraper
dos-2/oddshub	119	Terminal UI for odds analysis (Go)
sportsdataverse/oddsapiR	8	R wrapper for The Odds API

MCP Servers for Claude / AI Agents

Server	Purpose
Cloudbet Sports MCP	Live markets, odds, events from Cloudbet
Wagyu Sports MCP	Bridge to The Odds API
Apify Sports Analytics	12 tools: ESPN data, DFS projections, lineup optimization, 25+ leagues
FootballBin	Premier League + Champions League predictions
sports-leader-mcp	Free: scores, odds, injuries across 17 sports / 139 leagues
kitchenchem Odds API MCP	Odds comparison across bookmakers

Part 3 — The Real Playbook

Strategies Ranked by Real Win Rate

Strategy	Real Win Rate	Profit Source
Arbitrage	~100% per bet	Multi-book price mismatches (not prediction)
Bonus/Promo hunting	~99% EV	Free-bet conversion
Market-making (exchanges)	52-55%	Spread capture on Betfair/Polymarket
CLV edge vs Pinnacle close	54-58%	Beating Pinnacle's close by 2-5%
Line shopping at sharp books	+3-5% ROI	-105 vs -110 compounds
Reverse Line Movement	56-58%	Already in your system
Public Fade (NFL dogs)	58-64% ATS	Already in your system
Steam Following	57-60%	Already in your system

What Actually Wins Long-Term

1. **Beat Pinnacle's closing line by 2-5%** → 15-25% annual ROI
2. **Shop every bet across 6+ books** (Pinnacle, Circa, BookMaker, DK, FD, MGM)
3. **Bet early when lines open soft**, before sharp money shapes them
4. **Devig sharp lines** to find true probability — use goto_conversion or Crazy Ninja Odds devigger
5. **Use fractional Kelly ($\frac{1}{4}$ or $\frac{1}{2}$)** — never full Kelly; pros cap at 2.5% per bet
6. **Focus on liquid markets**: NFL spreads, NBA totals, MLB moneylines, Soccer Asian handicaps

Suggested Hook Additions for Your Repo

PreToolUse on Bash(settle) — auto-check CLV before settlement

PostToolUse on bet placement — log CLV to the tracker

SessionStart hook — fetch overnight line moves + steam alerts

Stop hook — daily P&L summary with CLV histogram

Honest Caveats

- Anyone promising "90%+ win rate" is selling a scam.
- Pinnacle, Circa, BookMaker, BetOnline tolerate winners. FanDuel / DraftKings / MGM **limit sharps fast**.
- The edge comes from **process** — line shopping, CLV tracking, bankroll discipline — not from one magic formula.

Appendix — Sources

- Cloudbet Sports MCP Server — https://cloudbet.github.io/wiki/en/docs/sports/api/mcp_integration/
- Awesome MCP Servers - Sports — <https://mcpservers.org/servers/apify-com-nexgendata-sports-mcp-server>
- Wagyu Sports MCP Market — <https://mcpmarket.com/server/wagyu-sports>
- CLV Guide 2026 (XCLSV) — <https://xclsvmedia.com/closing-line-value-clv-explained-the-complete-guide-for-sports-bettors-in-2026/>
- OddsJam CLV Tracker — <https://oddsjam.com/betting-education/closing-line-value>
- Unabated CLV Calculator — <https://unabated.com/betting-calculators/closing-line-value-calculator>
- Boyd's Bets - Beating CLV — <https://www.boydsbets.com/closing-line-value/>
- Sharp Football CLV Guide — <https://www.sharpfootballanalysis.com/sportsbook/clv-betting/>
- Kelly Criterion (Wikipedia) — https://en.wikipedia.org/wiki/Kelly_criterion
- OddsJam Kelly Calculator — <https://oddsjam.com/betting-calculators/kelly-criterion>
- betstamp Kelly Guide — <https://betstamp.com/education/kelly-criterion>
- Sports Betting Dime Kelly Guide — <https://www.sportsbettingdime.com/guides/strategy/kelly-criterion/>
- TipMaster Kelly Guide — <https://tipmaster.ai/blog/kelly-criterion-sports-betting-mathematical-edge>
- XCLSV Devigging Guide 2026 — <https://xclsvmedia.com/how-to-devig-betting-lines-in-2026-the-complete-guide-to-finding-true-odds/>
- XCLSV Line Shopping 2026 — <https://xclsvmedia.com/line-shopping-guide-2026-how-to-find-the-best-odds-and-save-thousands/>
- Pikkit - Sharpest Sportsbooks — <https://pikkit.com/blog/which-sportsbooks-are-sharp>
- Unabated - Market Makers — <https://unabated.com/articles/who-sets-the-sports-betting-line-market-makers>
- Bet Hero - Pinnacle as Sharp Ref — <https://betherosports.com/blog/how-to-use-pinnacle>
- PickTheOdds - Sharp Edges — <https://picktheodds.app/en/blog/sharp-sportsbooks-what-they-are-and-how-to-use-them-to-find-edges>